From Trained to Untrained: A Novel Change Detection Framework Using Randomly Initialized Models With Spatial–Channel Augmentation for Hyperspectral Images

Bin Yang, *Member, IEEE*, Yin Mao, Licheng Liu, *Member, IEEE*, Xinxin Liu, *Member, IEEE*, Yuzhong Ma, and Jing Li

***Abstract*— Deep learning (DL) approaches have been extensively applied to change detection in hyperspectral images (HSIs). However, the majority of them encounter scarcity of training samples or rely on complex structures and learning strategies. Although untrained change detection models have been proved to be effective in relieving the above problems, they were constructed using regular convolutions and treated spatial locations and channels equally, which are insufficient to extract discriminative features and lead to limited accuracy. Given this, a novel untrained framework using randomly initialized models with spatial–channel augmentation (RICD) is proposed for HSI change detection in this article. It consists of two major modules: 1) an enhanced feature extraction network using successive dilation-deformable feature extraction blocks, which can extract multiscale spatial–spectral features over unfixed sampling locations. It enlarges the field of view of convolutions and takes arbitrary neighborhood into consideration, which helps to increase the discriminativeness of the extracted features. And 2) a change-sensitive feature augmentation and comparison module integrating feature selection and spatial–channel augmentation strategies, which can exploit spatial context and channel importance. It magnifies difference between changed pixels and unchanged ones and emphasizes contribution of significant channels of the selected change-sensitive features. Despite that convolution operations are included in RICD, all the weights are untrained and fixed once they are randomly initialized, indicating that the RICD can work in an unsupervised manner. Its performance is tested over three widely used hyperspectral datasets. Quantitative and qualitative comparisons with several state-of-the-art unsupervised methods reveal the effectiveness of the RICD method.**

***Index Terms*— Change detection, deformable network, hyperspectral images, spatial–channel augmentation, untrained model.**

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Bin Yang, Yin Mao, Licheng Liu, and Xinxin Liu are with the College of Electrical and Information Engineering, Hunan University, Changsha 410082, China, and also with the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Changsha 410082, China (e-mail: binyang@hnu.edu.cn; maoyin@hnu.edu.cn; lichenghnu@gmail.com; liuxinxin@hnu.edu.cn).

Yuzhong Ma is with the Shandong Provincial Institute of Land Surveying and Mapping, Jinan 250102, China (e-mail: weixiao6316@163.com).

Jing Li is with the School of Information, Central University of Finance and Economics, Beijing 100081, China (e-mail: lijing2017@cufe.edu.cn).

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I. Introduction

LAND surface is always undergoing changes. Change detection aims to identify these changes accurately from remotely sensed images collected at different times [1]. It provides critical information on what is going on with land surface, and thus is widely used for land cover monitoring, disaster evaluation, decision-making, [2], [3], etc. Various types of remote sensing images can be used for change detection, such as multispectral, high resolution, synthetic aperture radar (SAR), and hyprespectral images (HSIs) [4]. Among these images, HSI is capable of recognizing characteristics of land surface objects with a fine spectral interval and over a wide range of wavelength, and thus is effective for improving the performance of change detection [5], [6].

Change detection in HSIs has been an active topic in the remote sensing community. A diverse range of methods have been proposed, which can be broadly classified into three categories, i.e., image algebra methods, image transformation methods, and image classification methods [7]. The image algebra methods perform subtraction, ratio, and other mathematical operations to compare the difference in multitemporal HSIs [8]. A popular algorithm is change vector analysis (CVA), which calculates the Euclidean distance for difference measurement [9]. And the compressed CVA method (C^2VA) [10] was developed from CVA for 2-D representation. These methods are simple and efficient but sensitive to radiation and noise due to lack of spatial information [11]. The image transformation methods, including PCA [12], MAD [13], and SFA [14], transform HSIs into new feature space for dimension reduction and change features emphasizing. However, it might be challenging to select appropriate thresholds to detect changes [15]. Image classification

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methods, i.e., postclassification [16] and compound classification [17], can measure classification results for obtaining multiple change information. Accordingly, accuracy is limited by the performance of classifiers [18]. Most of the above methods are originally proposed for multispectral images and are further applied or extended to HSIs [19]. They are generally designed based on the handcrafted features, and thus are limited for detecting complex change information.

Recently, thanks to the fast development of deep learning (DL) techniques [20], [21], the DL-based HSI change detection methods have immediately attracted tremendous attention [22]. The DL-based methods can extract discriminative features without handcrafted features, and thus are believed to be more intelligent for change detection [23], [24]. To improve the performance of DL-based HSI change detection methods, many attempts have been conducted. Diverse DL networks, e.g., deep belief networks (DBNs) [25], deep neural networks (DNNs) [26], and autoencoders (AEs) [27], have been applied on HSI change detection and proved to be efficient, but they mainly focus on spectral features. Moreover, massive works have explored convolution networks (CNNs) to represent spatial–spectral features. For example, a mixed-affinity matrix integrating unmixing was introduced to an end-to-end 2D CNN network (GETNET) for HSI change detection [28]. Zhan et al. [29] proposed a Siamese-CNN architecture trained by contrastive loss to extract spectral–spatial features from tensor pairs. To solve the information redundancy of HSIs, Ou et al. [30] presented a slow–fast band selection (SFBS) method in the CNN framework, involving feature grouping strategies to extract discriminative features. Gong et al. [31] and Wang et al. [32] embedded spatial and spectral attention mechanisms in the designed CNN networks to enhance feature representation, where informative channels and locations were emphasized and irrelevant spectral information and spatial information were suppressed. Taking account into temporal information, Song et al. [33] and Shi et al. [34] took advantages of CNN and long short-term memory (LSTM) to exploit spatial–spectral–temporal features, and multiclass change detection besides binary change detection experiments achieved gratifying results. While Wang et al. [35] successfully applied an end-to-end transformer model to extract spectral, spatial, and temporal sequence information for HSI change detection. Although these methods have achieved good results, most of them need massive manual labeled data to train the network for learning to extract deep features.

Notably, it is time-consuming and laborious to obtain manually labeled training samples for HSIs [36]. To alleviate this problem, some attempts have asked for methods with a small amount of labeled samples, e.g., Liu et al. [37] proposed a semi-supervised HSI change detection method that introduced active learning to select unlabeled samples as additional training sets for improving change detection performance. Qu et al [38] first introduced graph convolution networks (GCNs) for tackling HSI change detection with limited samples and embedded a difference amplification module in the dual-branch network to highlight change information. With the development of DL-based change detection technology, some unsupervised methods have been proposed to overcome the scarcity of manual labeled samples, which use pseudolabels generated from traditional unsupervised methods to train the designed deep models [26], [39]. For example, Shi et al [40] proposed a collaborative analysis framework trained by pseudolabels, in which a self-perception network and a cross-task knowledge transfer mechanism were designed to leverage the synergy of multiple datasets. Besides, Song and Kim [41] presented an unsupervised methods based on transfer learning, in which the model pretrained on source domain data are fine-tuned using generated labeled data for target images. There are methods that do not rely on pseudolabels, e.g., using adversarial loss to train the network in adversarial learning to conduct unsupervised change detection [42], and focusing on self-supervised DL framework for learning and clustering discriminative features to achieve unsupervised HSI change detection [43]. Nevertheless, these methods either are limited by the precision of generated pseudolabels or suffer from complicated network frameworks and learning strategies. This motivates us to explore whether DL models can be conveniently applied on HSI change detection without learning from labeled data.

It is encouraging that previous researches have provided both theoretical and practical bases for the assumption above, e.g., using the untrained models. First, the untrained DL models have proved to be effective and attracted increasing attention in recent years [44]. Despite labeled images are necessary for learning realistic image priors in DL, they are no longer needed for untrained models [45]. They were proposed based on the assumption that most of the existing neural networks are usually sufficient for capturing critical low-level features, even they are not trained from labeled data [46]. Given this, Lempitsky et al. [46] randomly initialized the network's weights and got remarkable performance in image denoising and super-resolution. Such untrained networks have been extensively explored for high resolution image processing tasks, e.g., phase imaging [47], face detection [48], and image recovery [49]. However, most of these algorithms update network parameters in iterative optimization using constraints of objective functions and thus encounter a problem of selection of the iteration number [45]. Moreover, an untrained convolution network with weights fixed after randomly initialization was proposed for image recognition and exhibited good performance, which avoided iterative optimization and reduced computational cost [50]. Second, the untrained network offers the opportunity to capture sufficient low-level semantic features from HSIs for accurate change detection. The reasons can be stated as follows. For one hand, HSIs have characteristics of relatively low spatial resolution, i.e., less spatial complexity than high resolution images. Thus, complex spatial information from deeply high-level feature representation for HSIs is of no great importance than that for high-resolution images [51]. Whereas the untrained networks have been used for processing high-resolution images [45], they are probable to represent spatial features from HSIs. For the other hand, HSIs reflect detailed information of ground

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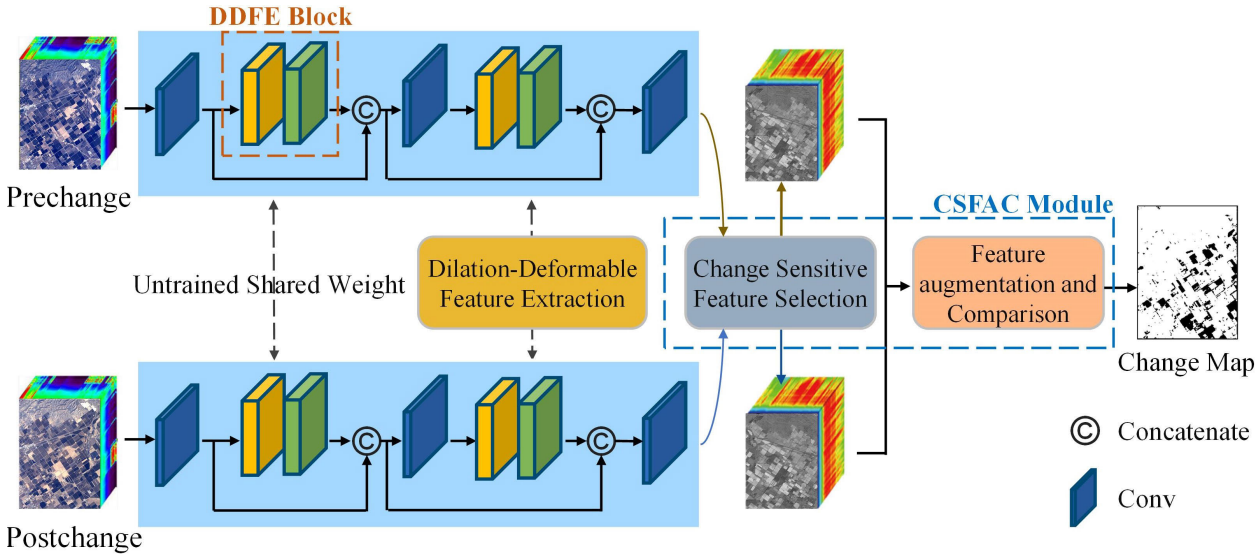


Fig. 1. Overview of the proposed RICD. DDFE: dilation-deformable feature extraction. CSFAC: change-sensitive feature augmentation and comparison.

objects from rich spectral features that can be extracted using untrained networks, which will be conducive to improve the performance of change detection [18]. Third, untrained network has been applied on the HSI change detection task and yielded remarkable results, in which a dual-branch untrained convolution network was designed to extract meaningful semantic features from bitemporal HSIs and combined deep CVA to distinguish between changed pixels and unchanged ones [51]. The proposed untrained network, which assigns weights fixed after initialized by the He initialization method, fully shows that the untrained model is sufficient to extract spatial–spectral information from HSIs. Besides, it has been further explored to fusing multiple untrained networks in the decision level to alleviate the variance resulted by turbulence of weights' initialization of the untrained models [52].

In view of the above analysis, untrained networks have great potential to break through the limitation of labeled samples for HSI change detection tasks. However, the existing implementations only use regular convolutions as feature extractors and treat spatial locations and channels equally, which are incompetent to deal with diverse scales of change and fully use abundant spatial and spectral information, leading to some erroneously detected regions [51], [52]. To tackle this, we propose a randomly initialized change detection framework (named as RICD) using untrained models with spatial–channel augmentation for HSIs. Different from the existing methods, a feature extraction network integrating dilation and deformable convolutions is designed in the RICD for powerful spatial–spectral local feature representation. Moreover, a deep feature augmentation and comparison strategy is proposed in terms of spatial context and channel importance, which can enhance discriminative features for better distinguishability and classify pixels represented by the local features into changed and unchanged types accurately. The main contributions of this article are as follows.

1) A novel randomly initialized framework named RICD is proposed for unsupervised HSI change detection, which integrates untrained DL techniques and feature augmentation strategies to achieve high accuracy without learning from labeled data. RICD can fully exploit hyperspectral properties and is effective in real application scenes.

2) A dual-branch dilation-deformable feature extraction (DDFE) network is built in the RICD for discriminative spatial–spectral feature representation. With successive DDFE blocks conducting convolutions in enlarged views and unfixed grid sampling locations, it can extract multiscale deformable features in untrained ways to tackling diverse scales' change.

3) A change-sensitive feature augmentation and comparison (CSFAC) module embeds feature selection and spatial–channel augmentation strategies to improve the discrimination ability of changes. The selected change-sensitive features are augmented in the light of spatial context and channel importance to magnify feature separability and emphasize contribution of informative channels.

II. Proposed Method

A. Overview

Fig. 1 illustrates the diagram of the proposed RICD method. Basically, it is an untrained framework consisting of a DDFE network and a CSFAC module. The DDFE network is responsible for extracting multiscale deformable spatial–spectral features from HSIs. While the CSFAC module aims to further enhance discriminative features from hyperdimensional features and provide the measurement of difference between bitemporal discriminative features to better distinguish change. The flowchart of RICD is generalized as follows.

First, given bitemporal HSIs as prechange image T_1 and postchange image T_2, both having a dimension of $H \times W \times C_\ell$. The designed weight shared dual-branch network is used to extract deep features from T_1 and T_2, respectively. This network has randomly initialized and fixed weights without learning from labeled data, thus is untrained and works in an unsupervised way. The key structure of the

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TABLE I
KEY STRUCTURE OF RICD. C_ℓ IS THE NUMBER OF INPUT CHANNELS, AND λ IS AN INTEGER ($\lambda \geq 1$)

Layer Name	Input Channel	Output Channel	Kernel size	Stride
Conv	C_ℓ	λC_ℓ	(3, 3)	1
DDFE	λC_ℓ	λC_ℓ	(3, 3)	1
Conv	$2\lambda C_\ell$	λC_ℓ	(3, 3)	1
DDFE	λC_ℓ	λC_ℓ	(3, 3)	1
Conv	$3\lambda C_\ell$	λC_ℓ	(3, 3)	1

network is presented in Table I. The first convolution layer is used for projecting T_1 and T_2 into high-dimensional feature space λC_ℓ, while the others for feature fusion at different levels and dimension reduction to reduce computation cost at the same time.

Then, DDFE blocks combining multi-dilation convolutions and deformable convolutions in the dual-branch network are used to extract multiscale features under an unfixed grid sampling strategy. In addition, the dense connection technology is applied between them to further enhance local features' propagation and extract more reliable spatial–spectral features.

After that, the CSFAC module conducts change-sensitive feature selection, feature augmentation, and comparison. Change-sensitive feature selection works as a strategy to obtain important features from hyperdimensional features for better detecting change.

Finally, feature augmentation applies spatial and channel augmentation on difference features from bitemporal deep features extracted above to further highlight the characteristics of change and increase separability between changed pixels and unchanged ones. This is followed by feature comparison, which uses a threshold for segmenting the distance image measured by a distance metric on augmented difference features to achieve the final change map (CM).

B. Dilation-Deformable Feature Extraction

In previous studies that use untrained networks for HSI change detection, only regular convolutions with fixed kernels have been conducted as feature extractors [51], [52], which overlooks diverse scales' change information in HSIs and may be insufficient for powerful feature representation. Thus, the DDFE block is designed in the proposed RICD to make full use of multiscale deformable spatial–spectral features. The structure of the DDFE block is presented in Fig. 2. It first uses two dilated convolution layers for extracting local features at diverse scales with different receptive fields. Then, the multi-dilation features are further put into a deformable convolution layer for deformable feature representation. With successive DDFE blocks, the proposed network can generate discriminative features of prechange image and postchange image from local representation at different levels for accurate change identification.

Compared with the traditional convolution, the dilated convolution can exponentially increase the receptive fields with only linear growth of parameter numbers [53]. It is thus a powerful tool for extracting local features in a relatively greater receptive field. Given input features

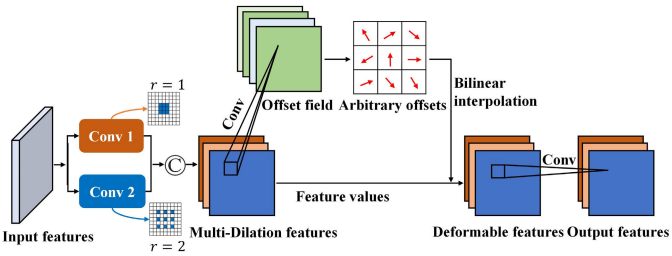


Fig. 2. DDFE block.

$X_t \in \mathbb{R}^{H_\ell \times W_\ell \times \lambda C_\ell} (t = 1, 2)$ of each branch, the dilated convolution operator is defined as

$$Y_t^r(X_t) = \sum_{c \in \lambda C_\ell} \theta_{k,r}^c * X_t^c \tag{1}$$

where $*$ is the convolution operator. The dilated convolution $\theta_{k,r}^c$ is conducted with a window size k and a dilation rate r. $Y_t^r(X_t)$ is the output features of one dilation layer from each branch with dilation rate r and kernel number of λC_ℓ, which also has the dimension of $\mathbb{R}^{H_\ell \times W_\ell \times \lambda C_\ell}$. With respect to the dilation rate r, the receptive fields of a window size k are enlarged to $k + (k-1)(r-1)$. It is notable that the dilated convolution corresponds to a regular convolution when $r = 1$. With increasing r, a larger receptive field is capable of extracting sufficient contextual information from the input. Taking into account the coarse spatial details of HSIs with relatively low spatial resolution, the dilation rates are set to 1 and 2 in this article, respectively, corresponding to 3×3 and 5×5 receptive fields.

Multiscale features extracted with the two r are fused in concatenation at channel dimension as multi-dilation features, which can be expressed as

$$\mathrm{MD}_t = \mathrm{Concat}\left(Y_t^1, Y_t^2\right) \tag{2}$$

where Concat$(\cdot)$ denotes the concatenation operation, and MD_t represents the features of T_1 and T_2 in each branch.

For the input features, the sample locations of a regular CNN are fixed over a 2-D window (with size $N \times N$) centered at p_0 and are denoted by

$$\mathcal{R} = \{(0,0), \quad (0,1), \ldots, (N-1, N-1)\}. \tag{3}$$

As shown in Fig. 3, the regular convolution only covers a regular box, while ignoring locations outside the box. These locations also contribute to accurate detection of changes as they convey surface structural information. The deformable convolution, which is proposed based on the idea of augmenting the sample locations with additional offsets [54], can break through the limitation of a regular CNN with fixed sampling locations. It has been proven efficient in remote sensing [55].

Inspired by this, and following the suggestions that deformable convolution is suitable for deeper layers [54], the DDFE block embeds a deformable convolution layer after multi-dilation convolutions to extract deformable features within an unfixed grid sampling structure, which enables the network to extract spatial–spectral features at a larger receptive field and over arbitrary neighbor locations. Two steps have been added in the deformable convolution: 1) random

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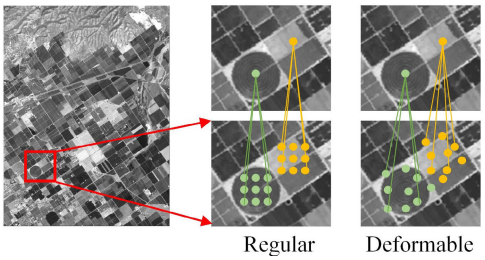


Fig. 3. Comparison between regular and deformable convolution. (Notably, the location offsets of deformable convolutions in the proposed RICD are randomly created.)

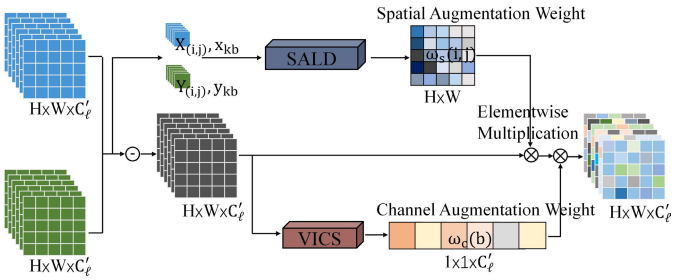


Fig. 4. Spatial–channel augmentation strategies. SALD: Spectral-angle-based local difference. VICS: Variance-importance-based change sensitivity.

creation of location offsets and 2) convolutional calculation of deformable features. As for the first step, the location offsets are generated using a regular convolution, which is initialized randomly by the He initialization method [56]. The channel dimension of the offset field is twice of the input feature size, corresponding to the offsets along two axes. As for the second step, the output dilation-deformable features $\mathrm{DD}_t(p_0)$ at location p_0 from the input features MD_t are formulated as

$$\mathrm{DD}_t(p_0) = \sum_{p_n \in \mathcal{R}} w(p_n) \cdot MD_t(p_0 + p_n + \triangle p_n) \tag{4}$$

where w denotes the weight of p_n enumerating the locations in $\mathcal{R}$. The location offsets are $\{\triangle p_n | n = 1, 2, \ldots, N\}$, where $N = |\mathcal{R}|$.

It is notable that the offsets $\triangle p_n$ are usually irregular and fractional. To project the offsets to integral spatial locations, bilinear interpolation operation is conducted as

$$\mathrm{MD}_t(p) = \sum_{q} \rho(q_x, p_x) \cdot \rho\big(q_y, p_y\big) \cdot MD_t(q) \tag{5}$$

where p represents the sum of p_0, p_n, and $\triangle p_n$, i.e., $p = p_0 + p_n + \triangle p_n$. q denotes the integral locations in the input features MD_t and $\rho(a, b) = \max(0, 1 - |a - b|)$.

C. Change-Sensitive Feature Augmentation and Comparison

The extracted hyperdimensional features from the DDFE network contain redundant information that is useless for change detection and may weaken the influence of significant features [18]. To solve this, change-sensitive feature selection block, feature augmentation, and comparison block are designed in the CSFAC module, which helps augment and compare the discriminative features from the outputs of the dual-branch DDFE network to accurately identify change.

The final feature pairs from the dual-branch network are denoted as $F_t \in \mathbb{R}^{H \times W \times \lambda C_\ell}$. Because the untrained weights are shared in the two branches, unchanged pixels should show similar feature representations, while changed pixels would be transformed into quite different feature representations in F_t.

First, the change-sensitive feature selection block serves to select the most significant channels from F_t that provide more change information, so as to better detect the change in low prior probability in change detection tasks, which uses a spatial-split-based and maximum-variance-based feature selection scheme, as documented in [51] and [57]. The extracted change-sensitive features from F_t are expressed as $F'_t \in \mathbb{R}^{H \times W \times C'_\ell}$.

Generally, a distance metric method is applied on F'_t to measure the deep feature difference for obtaining the final CM in an unsupervised way. In this article, the deep feature difference DI' is first calculated as

$$\mathrm{DI}' = |F'_2 - F'_1| \tag{6}$$

where $DI' \in \mathbb{R}^{H \times W \times C'_\ell}$.

Next, the spatial–channel augmentation strategy is conducted on DI' for discriminative features enhancement. To achieve enhanced feature representation without learning from labeled data, we conduct spatial and channel augmentation in novel ways that are different from most other advanced augmentation methods using convolution operation to realize attention mechanisms [58], as shown in Fig. 4. They are described as follows.

The spatial augmentation is proposed to further model the rich context features from local representation and increase the separability between changed pixels and unchanged ones. As shown in Fig. 4, we design a spectral-angle-based local difference (SALD) operation to measure the deep feature difference from F'_t, which can further exploit intrinsic local relationship from the neighborhood pixels. The SALD conducts nonlinear operation on the distance features calculated by the spectral angle weighted-based local absolute distance (SALA) method proposed in article [59] to obtain normalized difference features, indicating the change probability of each pixel. The differences between pixel pairs from F'_t are calculated in their neighborhoods. Given $X_{(i,j)} \in \mathbb{R}^{1 \times C'_\ell}$ and $Y_{(i,j)} \in \mathbb{R}^{1 \times C'_\ell}$ represent the pixels of F'_1 and F'_2 at location (i, j), respectively. The x_{kb} and y_{kb} are n neighboring pixels centered at $X_{(i,j)}$ and $Y_{(i,j)}$ in a single band b, respectively, where $k \in \{1, 2, \ldots, n\}$. The spatial augmentation weight $\omega_s \in \mathbb{R}^{H \times W}$ conducted though SALD operation is expressed as

$$\omega_s(i, j) = 1 - \exp\left(-\left(\sum_{k=1}^{n}\sum_{b=1}^{C'_\ell} |y_{kb} - x_{kb}|\right)\left(\frac{X_{(i,j)} Y_{(i,j)}}{|X_{(i,j)}\| Y_{(i,j)}|}\right)\right). \tag{7}$$

The spatial augmentation weight ω_s then can be conducted elementwise multiplication with DI' to enlarge the difference between changed pixels and reduce the difference between unchanged pixels.

The channel augmentation is introduced to further emphasize the contribution of important features and improve the adaptability in different scenarios. F'_t is selected by retaining a certain percentile of the sorted features [57]. For the

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reason that each channel from the deep feature difference DI' expresses different semantic information corresponding to ground projects in the scenarios, some irrelevant information may limit the change detection performance if each channel is treated equally [60]. Inspired by the method that using the variance for evaluating sensitivity to change information [57], we apply the ratio of standard deviation to quantify difference. We assume that channels with larger difference can contribute more reliable change information than that with less difference and propose a variance-importance-based change sensitivity (VICS) method to formulate the channel augmentation weight $\omega_c \in \mathbb{R}^{C'_\ell}$, which is defined as

$$\omega_c(b) = \frac{\text{std}\big(DI'(b)\big)}{\frac{1}{C'_\ell}\sum_{b\in C'_\ell}\text{std}(DI'(b))} \tag{8}$$

where std$(\cdot)$ represents the standard deviation operation of the feature values in a given channel. The ω_c thus transforms $H \times W \times C'_\ell$-dimensional features into C'_ℓ single values that convey change information of each channel. And a channel b with a larger $\omega_c(b)$ contains more reliable change information. The $\omega_c \in \mathbb{R}^{C'_\ell}$ is then reshaped to $\mathbb{R}^{1\times 1\times C'_\ell}$. The reshaped ω_c can be used for conducting elementwise multiplication with DI' to enhance significant channels which represent specified semantic information for better distinguishing change.

To integrate the spatial and channel augmentation weights, we perform elementwise multiplication operations on DI' with spatial augmentation weight and channel augmentation weight simultaneously to obtain the conjoint spatial–channel augmented deep feature difference, which is expressed as

$$\text{DI}'' = DI' \otimes \omega_s \otimes \omega_c \tag{9}$$

where $DI'' \in \mathbb{R}^{H\times W\times C'_\ell}$, and $\otimes$ represents the elementwise multiplication.

Finally, with the spatial–channel augmented deep feature difference, for a given pixel at location (i, j) in DI'', the deep feature comparison is further quantified using the Euclidean norm as

$$\text{DI}(i,j) = \sqrt{\sum_{b\in C'_\ell}\big\{\text{DI}''(i,j,b)\big\}^2}. \tag{10}$$

D. Initialization and Detection

In this section, we present the other details of the proposed RICD method.

First, we illustrate the initialization approach of the network. The proposed RICD method structures an untrained DL model, whose weights are fixed and merely initialized with a weight initialization technique to capture semantic features of HSIs with low-spatial and high-spectral resolution for change detection. There are some classical weight initialization strategies, e.g., Xavier initialization [61], He initialization [56], LSUV initialization [62], and Fixup initialization [63]. For the reason that He initialization is widely used and shows good performance, we use it to initialize the proposed network. It is conducted in two ways for the forward propagation and backward propagation process, respectively. During forward propagation, the variance of results of each convolution layer is set to 1, which is expressed as

$$\frac{1}{2}n_l\text{Var}[\omega_l] = 1, \quad \forall\, l \tag{11}$$

where n_l denotes the number of connections of a response for a convolution layer l, ω_l represents the random variables of each element in the filter weights W_l, and $Var[\cdot]$ is the variance calculation. During backward propagation, the variance of gradient of each layer is equal to 1, which is expressed as

$$\frac{1}{2}\hat{n}_l\text{Var}[\omega_l] = 1, \quad \forall\, l \tag{12}$$

where $\hat{n}_l$ denotes the number of connections of the gradient for a convolution layer. Thus, weights of the proposed network are initialized to a normal distribution with a mean of 0.

In terms of detection operation on DI to obtain the final CM, thresholding [26] or clustering techniques [64] are usually used to segment pixels into changed and unchanged types. The Otsu thresholding can automatically select the optimal threshold [65], which is simple, highly efficient, and is widely used in change detection tasks [66]. Thus, we apply the Otsu method on DI for segmentation to obtain the binary change result. It aims to maximize the separability of classes in gray levels to determine the optimal threshold. Assuming that pixels in DI are in L gray levels, the threshold at level t dichotomizes the pixels into two classes. The between-class variance $\sigma_b^2(t)$ is calculated as

$$\sigma_b^2(t) = \frac{\big[\mu_t z(t) - f(t)\big]}{z(t)[1-z(t)]} \tag{13}$$

where μ_t is the total mean gray level of DI, and $z(t)$ and $f(t)$ are the zero and first cumulative moments of the gray-level histogram of DI up to the kth level, respectively. Then the optimal threshold t^* consists with

$$\sigma_b^2\big(t^*\big) = \max_{1\le k\le L}\sigma_b^2(t). \tag{14}$$

With the threshold t^*, pixels $P(i, j)$ in DI are segmented as

$$R(i,j) = \begin{cases} 1, & P(i,j) > t^* \\ 0, & P(i,j) \le t^* \end{cases} \tag{15}$$

where $R(i, j)$ represents the pixel (i, j) in the CM, and "1" and "0" stand for "changed" and "unchanged," respectively.

The overall procedure of the proposed RICD method is illustrated in Algorithm 1.

III. Experimental Results

A. Hyperspectral Dataset Description

In this article, three widely used hyperspectral datasets were used for validation of the proposed method. Each dataset consists of bitemporal HSIs that have been preprocessed by atmospheric, geometric, and radiometric correction, and one ground-truth (GT) map with changed pixels colored black, unchanged pixels colored white, while unknown pixels colored gray. They are represented in detail as follows.

1) *Santa Barbara dataset*: Bitemporal images consist of 984 × 740 pixels with 224 spectral bands ranging from

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Algorithm 1 Overall procedure of the proposed RICD

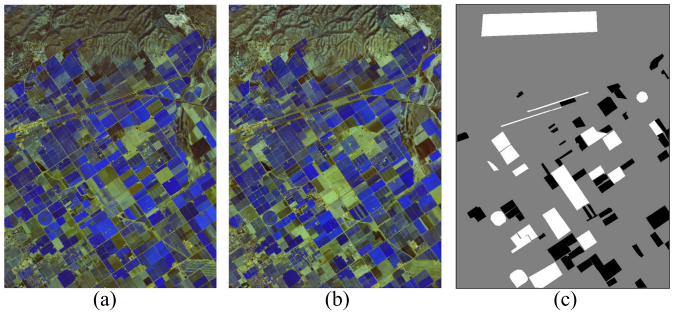
Input:

Prechange image T_1, Postchange image T_2

Output:

Change map R

1. **Initialization**: Randomly initialize the RICD with the He initialization technique and keep weights fixed;
2. Conv: Project T_t $(t = 1, 2)$ to X_t;
3. **DDFE block 1:**
   (1) Multi-dilation convolution:
   Extract features MD_t from X_t (1 and 2);
   (2) Deformable convolution: Extract features DD_t from MD_t (4);
4. Concatenate: $F_{1,t} = Concat(X_t, DD_t)$;
5. Conv: Project $F_{1,t}$ to $X_{2,t}$;
6. **DDFE block 2:**
   (1) Multi-dilation convolution:
   Extract features $MD_{2,t}$ from $X_{2,t}$ (1 and 2);
   (2) Deformable convolution: Extract features $DD_{2,t}$ from $MD_{2,t}$ (4);
7. Concatenate: $F_{2,t} = Concat(X_t, DD_t, DD_{2,t})$;
8. Conv: Project $F_{2,t}$ to F_t;
9. **Change-sensitive feature selection:** calculate F_t' from F_t;
10. Deep feature difference: obtain DI' (Eq.6);
11. **Spatial–channel augmentation:** Calculate ω_s and ω_c (7 and 8);
12. Augmented deep feature difference: Calculate DI'' (9);
13. **Deep feature comparison:** Obtain DI by Euclidean norm (10);
14. Calculate $t*$ using Otsu method (14);
15. Obtain change map R (15).

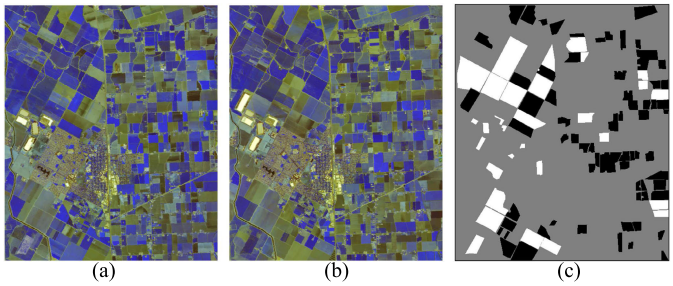


(a) (b) (c)

Fig. 5. Santa Barbara dataset. False color images (R: band 50, G: band 20, B: band 10). (a) Prechange image; (b) Postchange image; and (c) GT map (white: unchanged, black: changed, gray: unknown).

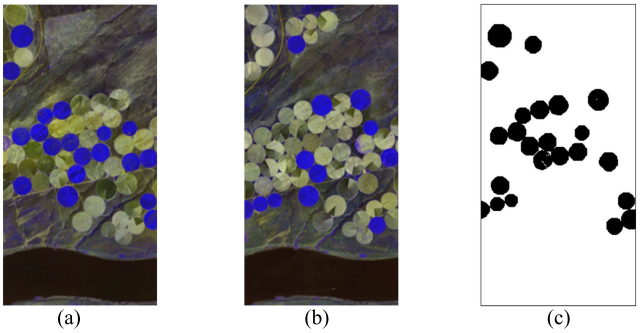
0.4 to 2.5 μm, as shown in Fig. 5. They were collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in the year of 2013 and 2014 over Santa Barbara region in California, United States. The two HSIs have a spatial resolution of 20 m and mainly describe sceneries about mountainous and farmland. A total of 132 552 pixels are referenced as GT, 52 134 of which are changed and 80,418 of which are unchanged.

2) *Bay Area dataset*: Bitemporal images are composed of 600 × 500 pixels with 224 spectral bands ranging from 0.4 to 2.5 μm, as shown in Fig. 6. They were collected by the



(a) (b) (c)

Fig. 6. Bay Area dataset. False color images (R: band 50, G: band 20, B: band 10). (a) prechange image; (b) postchange image; and (c) GT map (white: unchanged, black: changed, gray: unknown).



(a) (b) (c)

Fig. 7. Hermiston dataset. False color images (R: band 52, G: band 31, B: band 22): (a) prechange image; (b) postchange image; and (c) GT map (white: unchanged, black: changed).

AVIRIS sensor in the year of 2013 and 2015 over Patterson City area in California, United States. The scenes described in the two HSIs are mostly urban areas and farmland. There are 73 481 pixels referenced as GT, 39 270 of which are changed and 34 211 of which are unchanged.

3) *Hermiston dataset*: Bitemporal images have 390 × 200 pixels with 242 spectral bands ranging from 0.4 to 2.5 μm, as shown in Fig. 7. They were collected by the Earth Observing-1 (EO-1) Hyperion sensor in the year of 2004 and 2007 over Hermiston City area in Oregon, United States. Among 242 bands, we only used the remaining 198 bands with bands B001–B007, B058–B076, and B225–B242 excluded for not calibrated. The two HSIs have a spatial resolution of 30 m and are related to dynamic in crop fields. All the pixels are referenced as GT, 9986 of which are changed while the remaining are unchanged.

B. Experiment Description

The proposed unsupervised RICD method structures an untrained model that works in fixed randomly initialized weights without learning from labeled data. To evaluate its performance, four competitive unsupervised changed detection methods, including CVA [9], SFA [14], DCVA [57], and UHM-DCVA [51], are selected for comparison, which are sketched as follows.

(1) *Change Vector Analysis:* CVA is a classical unsupervised method which calculates spectral change vectors' difference in algebra methods and uses a confidence threshold for segmentation to acquire change results [9].

(2) *Slow Feature Analysis (SFA):* SFA is a transformation-based method that extracts slowly varying features from input images in the new space for identifying change, where

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invariant components are suppressed and changed pixels are highlighted for better separated [14].

(3) *Deep CVA (DCVA):* DCVA is a method that introduces transfer-learning-based networks for change detection. It exploits a multilayered CNN pretrained on other datasets for multiple scaled feature extraction, and CVA method is integrated for deep feature comparison [57].

(4) *Untrained Hyperdimensional Multiple DCVA (UHM-DCVA):* UHM-DCVA proved the untrained deep model to be effective in change detection for hyperdimensional images. The model is initialized with a weight initialization strategy for useful semantic features extraction from bitemporal images, while features are compared in DCVA to distinguish between changed pixels and unchanged ones [51].

It is notable that despite there are convolutional layers (i.e., regular convolution, dilated convolution, and deformable convolution), the proposed model is untrained and works in an unsupervised way without labeled data for learning. All the weights are untrained while fixed after only initialized using the He initialization method, in which the weights are mutually independent but share the same distribution [56].

The performances of these methods are quantified using five indices, i.e., precision (Precision), recall (Recall), overall accuracy (OA), Kappa coefficient (Kappa), and F1 score. Precision describes how many of the changed pixels detected by the method are accurate. Recall represents how many of the actual changed pixels have been correctly detected. OA provides overall accuracy calculating the proportion of correctly detected pixels to overall pixels. Kappa statistically measures the consistency between the CM and GT map. F1 score defines a balanced index that can be considered as harmonic mean of Precision and Recall. They are calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{16}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{17}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{18}$$

$$\text{PRE} = \frac{(\text{TP}+\text{FP}) \times (\text{TP}+\text{FN}) + (\text{TN}+\text{FN}) \times (\text{TN}+\text{FP})}{(\text{TP}+\text{FP}+\text{TN}+\text{FN})^2} \tag{19}$$

$$\text{Kappa} = \frac{\text{OA} - \text{PRE}}{1 - \text{PRE}} \tag{20}$$

$$F1 \text{ score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{21}$$

where TP, TN, FP, and FN separately indicate the numbers of pixels correctly detected as changed, correctly detected as unchanged, erroneously detected as changed, and erroneously detected as unchanged.

To relief the influence of randomness in initialization, performances of UHM-DCVA and RICD are calculated as the mean value of five runs.

C. Setting Details

The proposed RICD has flexible structures by varying the number of channels in Table I and the number of DDFE blocks in Fig. 1, which may have major impact on change detection results through affecting the feature extraction process. Additional experiments were performed on the Santa Barbara dataset to make the best setting for the proposed method.

TABLE II
ACCURACY WITH DIFFERENT NUMBERS OF CHANNELS (%)

Channel number	Precision	Recall	OA	Kappa	F1 score
$\lambda = 1$	94.98	91.91	94.90	89.24	93.41
$\lambda = 2$	**96.30**	90.66	94.96	89.32	93.39
$\lambda = 3$	96.27	**92.30**	**95.56**	**90.63**	**94.24**
$\lambda = 4$	96.16	91.73	95.31	90.08	93.89

1) *Influence of the Number of Channels*: The number of channels in intermediate layers is important for the accuracy of the proposed RICD method. Here, the performance of RICD is tested with different numbers of channels in intermediate layers by varying the value of λ ($\lambda = 1, 2, 3, 4$) using the Santa Barbara dataset, as shown in Table II. As one can see, the best performance was achieved when $\lambda = 3$ in terms of almost all the indices. It can be explained as follows. With the increase in λ, more untrained weights are initialized for feature extraction, which helps increase the probability of generating discriminative features and thus increases the accuracy of the RICD method. However, once the number of channels is great enough, the increase in λ might also increase the occasionality of unreliable feature generation, thus decreases the accuracy of the feature selection scheme. Table II shows that $\lambda = 3$ is a good balance between accuracy and occasionality and is thus selected in this article.

2) *Influence of the Number of DDFE Blocks*: In the proposed RICD method, we designed the DDFE block to extract multiscale features from unfixed gridding sampling regions, thus enhancing the local feature representation ability. This block is easy to change in quantitative terms and can be conveniently embedded in other methods. While the number of DDFE blocks impacts the performance of the proposed RICD method. Fig. 1 illustrates the structure of RICD with two blocks. Such structure can be revised with fewer or more blocks. Table III presents the performance of RICD with different numbers of DDFE blocks using the Santa Barbara dataset (block number = 1, 2, 3). It is notable that the dense connection strategy was applied to each case. As can be seen from Table III, RICD with two blocks performs the best. It provides much better performance compared with that with only one block. The reason is that more blocks help extract more reliable local features, which results in better comparison results. However, with the increase in the number of blocks, besides the high cost of computation, the receptive field is too large to extract reliable local features because of the relatively less spatial complexity of HSIs. The performance of RICD with three blocks is thus less accurate than that with two blocks. Given this, the proposed RICD is constructed with two blocks.

D. Performance Comparison

The qualitative visualization results and quantitative objective indices for comparative methods on three datasets are elaborated in the following.

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TABLE III
ACCURACY WITH DIFFERENT NUMBERS OF DDFE BLOCKS (%)

Block number	Precision	Recall	OA	Kappa	F1 score
1	**96.83**	89.11	94.57	88.45	92.80
2	96.27	92.30	**95.56**	**90.63**	**94.24**
3	95.92	**92.53**	95.51	90.52	94.18

1) *Experiment on the Santa Barbara Dataset:* Qualitative change detection results of comparison methods on the Santa Barbara dataset are shown in Fig. 8. As can be seen from this figure, the SFA method achieves the most "noisy" CM with some scattered unchanged pixels misclassified as changed pixels. This phenomenon is largely attributed to less consideration of spatial information among adjacent pixels, and thus scattered pixels exist. As one of the most classical change detection methods, the CVA method has similar results with the GT map, while there are still noisy points due to lack of spatial information. With enhanced feature extraction ability provided by convolutional operations, there is less scattered noise in the CM of the DCVA method. However, the model pretrained on aerial data probably did not fit the hyperspectral dataset well as domain gap exists, leading to some missed real changed regions and pseudodetected changes. The UHM-DCVA's and RICD's results are much accurate than the above. For the UHM-DCVA method, the designed model with weights randomly initialized is sufficient to capture deep image prior, and the feature selection scheme helps extract more discriminative information, thus significantly improved the change detection performance. As for the proposed RICD method, its performance is even better, the DDFE and change-sensitive feature augmentation blocks contribute the untrained model to detect more completed change regions with even less false detection.

From Table IV, it is concluded that the DCVA method has the lowest OA and other indices for the Santa Barbara dataset, and the performance of the SFA method has improved by 2.60%–6.11% for the five indices compared with the DCVA method. The CVA method has slightly improved results, but the values of five indices except Precision are still less than 90%. Furthermore, the UHM-DCVA method obtains relatively high performance with the values of Precision, OA, and F1 score being 98.13%, 93.59%, and 91.28%, respectively. The proposed RICD method gets the highest indices except Precision value which decreases by 1.74% than the UHM-DCVA method. Moreover, the Recall value of the RICD method has increased by 6.7% than the UHM-DCVA method, which means that more real changes have been detected.

2) *Experiment on the Bay Area Dataset:* The SFA method erroneously identifies a large area relevant to farmland at the lower left as changed, and more changed pixels in the right area are categorized as unchanged, as shown in Fig. 9. The false detection and missed detection of change in the SFA method are effectively inhibited for the CVA method though both of them encounter spots in the CMs. For untrained DL-based methods, the DCVA method has false detections in some unchanged areas and broken boundaries in changed areas, indicating that transferred DCVA model is less robust

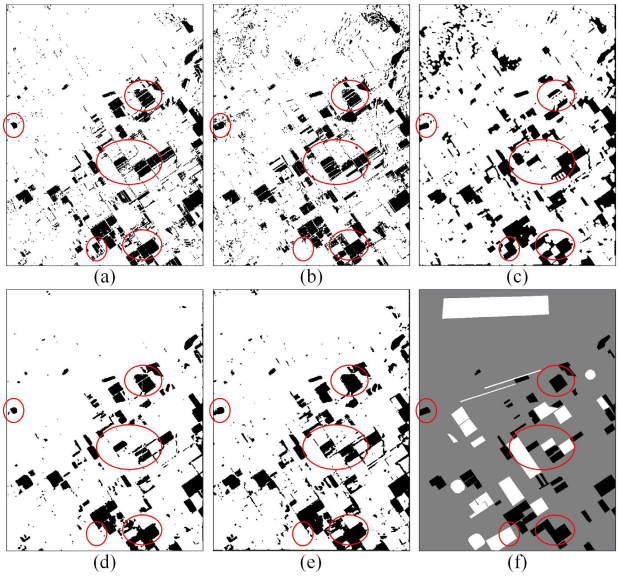


Fig. 8. Qualitative results of different methods on the Santa Barbara dataset (white: unchanged, black: changed, major differences are circled red). (a) CVA. (b) SFA. (c) DCVA. (d) UHM-DCVA. (e) RICD. (f) GT map.

to different scenes. Compared with the methods above, the UHM-DCVA method has relatively clean results with some isolated noise removed. Besides, a little missed change mainly appears at the right part. It is obvious that the RICD method achieves the most similar results to the GT maps. The superior performance is credited to more comprehensive exploration of change concerned space spectrum information and increase in separability between different classes in the proposed method.

The quantitative evaluation of comparison methods on the Bay Area dataset is presented in Table IV. The OA of the SFA method is lower than 80%, leading to the worst performance. With Recall and Kappa being 69.78% and 55.07%, respectively, the SFA method has high miss alarm rate and is not feasible to cope with class imbalance in change detection for HSIs. The results of the DCVA and CVA methods are gradually improved. Their comprehensive indices including OA and F1 score exceed 84% though Kappa around 70% is relatively low. By designed network architecture for HSIs and feature selection strategy, the UHM-DCVA method can work satisfactorily and it yields OA higher than 90% and other indices higher than 80%. Moreover, it has the highest Precision among all the methods. While the RICD method achieves the best performance though Precision is 0.05% lower than that of UHM-DCVA. The Recall of 90.52% and Kappa of 86.76% are greatly ameliorated compared with UHM-DCVA's 84.34% and 80.51%, getting fewer miss detections while keeping high OA. It demonstrates the effectiveness of our feature extraction and augmentation strategies for detecting change with low prior probability accurately and constricting class-imbalance.

3) *Experiment on the Hermiston Dataset:* Fig. 10 shows the visual results of five methods on the Hermiston dataset. The change regions in the Hermiston scene are mostly round irrigated agricultural fields. As can be seen, a lot of details are lost in DCVA's CM, especially the changed edges have serious expansion. In addition, an unchanged area in the lower

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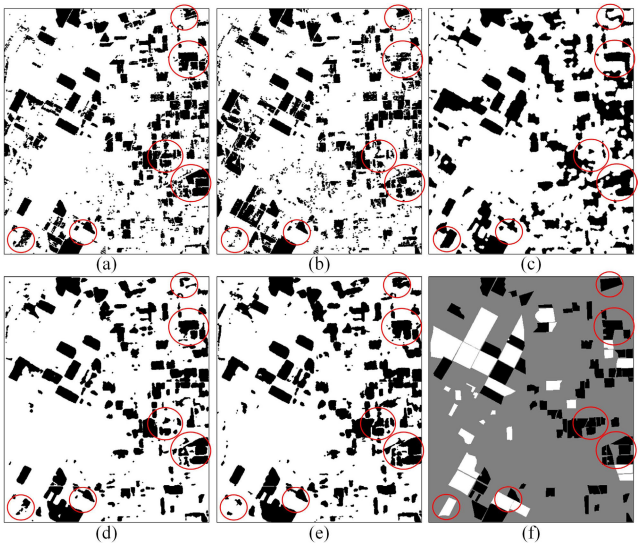


Fig. 9. Qualitative results of different methods on the Bay Area dataset (white: unchanged, black: changed, major differences are circled red). (a) CVA. (b) SFA. (c) DCVA. (d) UHM-DCVA. (e) RICD. (f) GT map.

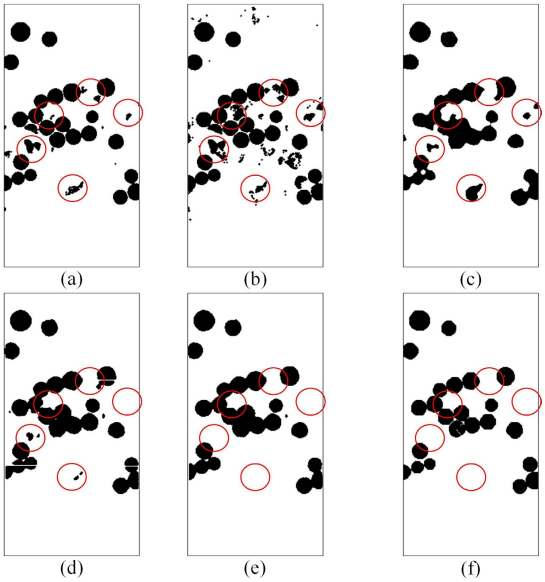


Fig. 10. Qualitative results of different methods on the Hermiston dataset (white: unchanged, black: changed, major differences are circled red). (a) CVA. (b) SFA. (c) DCVA. (d) UHM-DCVA. (e) RICD. (f) GT map.

middle of result map is apparently misclassified as changed. As for the SFA method, the real change regions and their boundaries are fit well in the CM. Nevertheless, this method has higher false alarm rate, as shown in the middle area of Fig. 10(a). Although having better CM compared with the SFA and DCVA methods, the CVA method also has massive false detections mainly in the left-middle part. In CMs of the RICD and UHM-DCVA method, false detections illustrated in the CVA and SFA methods have been improved to a great extent. Meanwhile, the expansion of change edges due to convolutional operations in the DCVA method is attenuated with enhanced feature extraction capability in these methods. On the whole, the RICD method gains the closest CM to the GT, indicating the superior performance among these methods.

Table IV presents the indices describing the change detection accuracy of methods on the Hermiston dataset. The five methods, i.e., SFA, DCVA, CVA, UHM-DCVA, and RICD, have relatively close OA of 94.81%, 96.57%, 97.41%, 97.86%, and 98.18%, severally. The reason is that change regions are in small proportion for the Hermiston dataset and all methods have correctly detected most of the scene. According to Table IV, the SFA method has the lowest Precision of 73.19%, leading to massive false detections. The DCVA method has the lowest Recall of 92.13%, and thus missed detection is the most acute. Compared with the SFA and DCVA methods, the CVA method can distinguish each ground object more precisely with Precision, Recall, and Kappa improved significantly. The UHM-DCVA method has Recall 3.34% higher and Kappa 2.07% higher than those of the CVA method, thus greatly improved the performance. While the RICD method is superior to the others, with all the indices improved between 0.32% and 1.59% compared with the UHM-DCVA method. From both qualitative and quantitative perspective, the proposed method shows superiority in change detection for HSIs.

TABLE IV
QUANTITATIVE CHANGE DETECTION RESULTS

Dataset	Accuracy(%)	CVA	SFA	DCVA	UHM-DCVA	RICD
Santa Barbara	Precision	90.61	86.50	81.68	**98.13**	96.27
	Recall	80.10	80.17	77.57	85.34	**92.30**
	OA	88.91	87.28	84.33	93.59	**95.56**
	Kappa	76.28	72.99	66.88	86.26	**90.63**
	F1 score	85.03	83.21	79.57	91.28	**94.24**
Bay Area	Precision	95.72	85.14	91.77	**96.95**	96.90
	Recall	78.93	69.78	78.05	84.34	**90.52**
	OA	86.86	77.34	84.53	90.21	**93.38**
	Kappa	73.93	55.07	69.25	80.51	**86.76**
	F1 score	86.52	76.70	84.36	90.20	**93.59**
Hermiston	Precision	87.36	73.19	82.97	87.88	**89.47**
	Recall	93.27	93.83	92.13	96.61	**97.28**
	OA	97.41	94.81	96.57	97.86	**98.18**
	Kappa	88.73	79.25	85.33	90.80	**92.15**
	F1 score	90.22	82.23	87.31	92.03	**93.20**

E. Ablation Analysis

To verify the effectiveness of the proposed RICD method, we implemented an ablation study in which the hyperparameters were set according to Section III-C.

1) *Dilated Convolution:* In the proposed RICD, we use two dilated convolution layers in the DDFE blocks for extracting local features at diverse scales with different receptive fields, and at the expense of lessened computational load. To evaluate their effect, ablation analysis is performed on experiments with different dilation rates. As shown in Table V, compared with regular convolutions (dilation rates are 1 and 1), dilated convolutions provide an advantage in terms of extracting multiscale information for improved change detection results. And the best performance is achieved by two parallel dilated convolution layers with dilation rates of 1 and 2 (RICD). Three parallel dilated convolution layers with dilation rates of 1, 2 and 3 obtain inferior outcomes, demonstrating that larger receptive fields are unsuitable for HSIs with less spatial complexity. Therefore, dilated convolutions with dilation rates of 1 and 2 can take account into the benefits of

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TABLE V

INFLUENCE OF THE DILATED CONVOLUTIONS (%)."KS" AND "DR" MEAN "KERNEL SIZE" AND "DILATION RATE" OF DILATED CONVOLUTION LAYERS, RESPECTIVELY

Dataset	KS	DR	Precision	Recall	OA	Kappa	F1 score
Santa Barbara	[3,3]	[1,2]	96.27	**92.30**	**95.56**	**90.63**	**94.24**
	[3,3,3]	[1,2,3]	**96.90**	91.56	95.52	90.53	94.14
	[3,5]	[1,1]	96.32	91.52	95.29	90.05	93.86
Bay Area	[3,3]	[1,2]	**96.90**	**90.52**	**93.38**	**86.76**	**93.59**
	[3,3,3]	[1,2,3]	96.01	90.12	92.71	85.43	92.97
	[3,5]	[1,1]	96.15	88.84	92.13	84.28	92.35
Hermiston	[3,3]	[1,2]	**89.47**	97.28	**98.18**	**92.15**	**93.20**
	[3,3,3]	[1,2,3]	87.75	**97.35**	97.92	91.10	92.30
	[3,5]	[1,1]	89.37	97.05	98.14	91.98	93.05

TABLE VI

INFLUENCE OF THE DEFORMABLE CONVOLUTIONS (%). "REGULAR" MEANS RICD USING REGULAR CONVOLUTIONS WHILE WITHOUT DEFORMABLE CONVOLUTIONS

Dataset	Method	Precision	Recall	OA	Kappa	F1 score
Santa Barbara	REGULAR	96.18	91.45	95.21	89.87	93.75
	RICD	**96.27**	**92.30**	**95.56**	**90.63**	**94.24**
Bay Area	REGULAR	96.89	89.29	92.74	85.49	92.93
	RICD	**96.90**	**90.52**	**93.38**	**86.76**	**93.59**
Hermiston	REGULAR	**90.18**	97.24	**98.29**	**92.59**	**93.58**
	RICD	89.47	**97.28**	98.18	92.15	93.20

dilated convolutions and hyperspectral properties to boost the efficiency of the proposed RICD.

2) *Deformable Convolution:* The deformable convolution layers designed for RICD use the unfixed grid sampling strategy to extract discriminative local features, which surmounts the constraint of regular CNNs. Here, ablation experiments were carried out on the three datasets to quantitatively and intuitively explore the validity of deformable convolutions for change detection in HSIs. Instead, all the deformable convolution layers in the proposed RICD were replaced by regular convolution layers ("REGULAR" in Table VI). Table VI gives the comparison results. With deformable convolutions, five indices, i.e., Precision, Recall, OA, Kappa, and F1 score, increase by 0.09%, 0.85%, 0.35%, 0.79%, and 0.49% for the Santa Barbara dataset, 0.01%, 1.23%, 0.64%, 1.27%, and 0.66% for the Bay Area dataset, and −0.71%, 0.04%, −0.11%, −0.44%, and −0.38% for the Hermiston dataset, respectively. For the first two datasets, the deformable convolutions contribute a lot to change detection accuracy, especially for Recall, Kappa, and F1 score, which means that RICD with deformable convolutions is more precious and robust to the change class. As for the Hermiston dataset, deformable convolution operations are possible to introduce futile context around boundaries of circular change areas to interfere feature extraction, thus leading to slightly decreased indices imputed to the expansion of change edges. Taking account of the more complex circumstances and significant improvements in the Santa Barbara and Bay Area datasets, it is convincing that deformable convolutions are profitable for change detection in HSIs.

3) *Spatial–Channel Augmentation:* In the CSFAC module, two weight strategies, i.e., spatial augmentation weight and channel augmentation weight, are proposed to enhance the feature representation. The spatial augmentation weight can further integrate local features to increase the separability between changed pixels and unchanged ones. The channel augmentation weight can emphasize the contribution of important features that are useful for detecting change while weakening the influence of unimportant features. To evaluate the effectiveness of the two weight strategies, comparisons were conducted over three datasets, respectively. We implemented this study in three ways, in which "w/o W" means no weight strategies were used, "w/s W" means only spatial augmentation weight was used, "w/c W" means only channel augmentation weight was used, while "RICD" used both the weights. Table VII shows the quantitative results. As can be seen, the other four evaluation indices except Precision increase in most situations for "w/s W" compared with the results of "w/o W" while decrease for "w/c W" over "RICD," which demonstrates the performance advantage of the spatial augmentation based on the SALD operation. With visualization displayed in Fig. 11 close to the GT map, the spatial augmentation weight provides reliable change probability of each pixel, where changed pixels have a greater weight, while unchanged pixels have the opposite. It acts on the deep feature difference and can further accurately magnify the difference between changed and unchanged pixels to improve change detection performance. Despite that the Precision or Recall may increase or decrease, the other three overall evaluation indices all increase for "w/c W" compared with the results of "w/o W," indicating the efficiency of the channel augmentation weight. In the experiment of "RICD," indices further upgrade for the Santa Barbara and Bay Area datasets though descend a little for the Hermiston dataset than those of "w/c W." Notably, the results of "RICD" outperform those of "w/o W." As a whole, these experiments represent that both the augmentation weights proposed in the RICD method are conducive to boost the accuracy of change detection.

TABLE VII

INFLUENCE OF THE AUGMENTATION WEIGHT STRATEGIES (%). "W/O W" MEANS STRATEGY WITHOUT AUGMENTATION WEIGHT. "W/S W" MEANS STRATEGY WITH SPATIAL AUGMENTATION WEIGHT. "W/C W" MEANS STRATEGY WITH CHANNEL AUGMENTATION WEIGHT

Dataset	Method	Precision	Recall	OA	Kappa	F1 score
Santa Barbara	w/o W	**96.96**	91.20	95.41	90.29	93.99
	w/s W	96.89	91.33	95.43	90.34	94.02
	w/c W	96.39	92.04	95.51	90.52	94.16
	RICD	96.27	**92.30**	**95.56**	**90.63**	**94.24**
Bay Area	w/o W	96.70	90.60	93.32	86.64	93.55
	w/s W	96.68	**90.68**	93.35	86.70	93.58
	w/c W	**96.92**	90.44	93.35	86.70	93.56
	RICD	96.90	90.52	**93.38**	**86.76**	**93.59**
Hermiston	w/o W	88.03	97.90	98.02	91.54	92.68
	w/s W	87.86	**97.96**	98.00	91.46	92.61
	w/c W	**89.55**	97.24	**98.19**	**92.18**	**93.22**
	RICD	89.47	97.28	98.18	92.15	93.20

4) *Initialization Strategy:* In the proposed RICD framework, the DDFE network is given a random initialization with the He initialization (named as HeInit) [56] strategy to perform as effective feature extractors. Table VIII shows the results of experiments investigating RICD with other initialization

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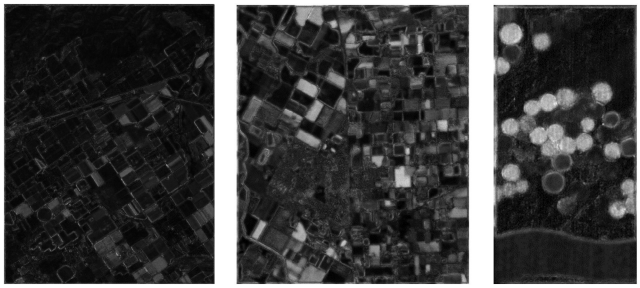


Fig. 11. Visualization of spatial augmentation weight on three datasets (from left to right: Santa Barbara dataset, Bay Area dataset, Hermiston dataset).

TABLE VIII
INFLUENCE OF THE INITIALIZATION STRATEGIES (%)

Dataset	Method	Precision	Recall	OA	Kappa	F1 score
Santa Barbara	HeInit	**96.27**	92.30	**95.56**	**90.63**	**94.24**
	Xavier	90.00	83.73	89.94	78.65	86.74
	LSUV	89.62	83.16	89.58	77.89	86.25
	Fixup	88.64	**95.06**	93.25	86.05	91.73
	OrthoInit	91.12	83.38	90.27	79.30	87.08
Bay Area	HeInit	**96.90**	90.52	**93.38**	**86.76**	**93.59**
	Xavier	93.82	87.65	90.32	80.64	90.63
	LSUV	93.80	85.49	89.22	78.49	89.44
	Fixup	90.39	**92.16**	90.57	81.02	91.26
	OrthoInit	94.14	86.34	89.82	79.67	90.06
Hermiston	HeInit	89.47	97.28	98.18	92.15	93.20
	Xavier	92.13	94.26	98.23	92.17	93.18
	LSUV	**93.61**	94.24	**98.44**	**93.02**	**93.92**
	Fixup	63.18	**99.31**	92.44	72.92	77.17
	OrthoInit	92.98	94.33	98.36	92.70	93.64

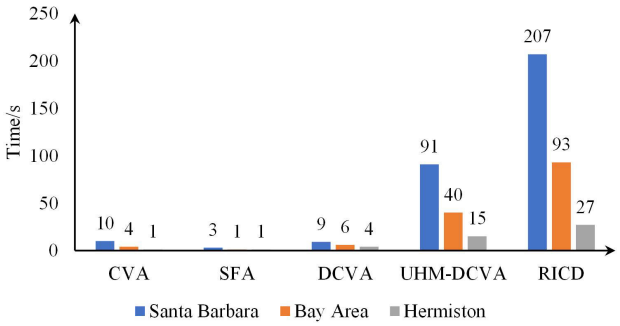


Fig. 12. Time consumption of methods on three datasets.

strategies, i.e., Xavier [61], LSUV [62], Fixup [63], and orthogonal initialization (named as OrthoInit) [67]. Compared with the majority methods in Table IV, RICD with various initialization strategies obtain improved results, indicating its effectiveness. The robust He initialization strategy that particularly takes account into rectifier nonlinearities can facilitate discriminative features extraction and achieve the best performance in most situations. It is thus selected in this article.

5) *Time Consumption:* In Fig. 12, the time consumption of the proposed RICD and comparison methods on three datasets is presented. It is clear that the CVA, SFA, and DCVA methods obtain the most efficient results for the reason they do not require training and initializing procedures. Both the UHM-DCVA and RICD methods use randomly initialized frameworks with untrained models and thus take longer time. The time required for RICD is higher than that of UHM-DCVA because of additional computational complexity of deformable convolutions and the spatial–channel augmentation. However, the time consumption is considered acceptable since majority HSI change detection tasks are offline and it can further be reduced when powerful machine is used for calculation. On the contrary, accuracy is the priority of these tasks. Consequently, we concentrate on the model's accuracy at the sacrifice of a little bit more time consumption.

IV. CONCLUSION

This article represents a novel framework using randomly initialized models with spatial–channel augmentation for change detection in HSIs, i.e., the aforementioned RICD method. The proposed RICD works in fixed weights without training to learn from labeled data. It is able to predict changed and unchanged pixels using the DDFE network and CSFAC module with full use of abundant information in HSIs. The DDFE network serves as enhanced feature extractors to achieve multiscale deformable spatial–spectral features, while feature selection and augmentation strategies in the CSFAC module are responsible for enhancing local feature representation and increasing distinguishability between changed and unchanged pixels. The major advantages of the RICD method are, on one hand, it can extract augmented discriminative local features with larger receptive fields and over unfixed sampling locations, making it more sensitive to structural information and change; on the other hand, although several convolution operations are conducted, it can work in an unsupervised manner. Experiments indicate that the RICD is superior to several other unsupervised methods. The RICD works because HSIs with relatively low spatial resolution have less spatial complexity compared with high-resolution images. It is notable that RICD is only a complement, but not a competitor of the current supervised DL-based methods. Whether it can be directly applied to change detection in high-resolution images remains uncertain and needs further studies.

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Bin Yang (Member, IEEE) received the B.S. degree in science and technology of remote sensing from Beihang University, Beijing, China, in 2012, and the Ph.D. degree in photogrammetry and remote sensing from Peking University, Beijing, in 2017.

He is currently an Associate Professor with the College of Electrical and Information Engineering, Hunan University, Changsha, China, and with the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Changsha. His research interests include remote sensing and artificial intelligence.



Yin Mao received the B.S. degree in automation from Xinjiang University, Ürümqi, China, in 2020. She is currently pursuing the M.S. degree in control science and engineering with Hunan University, Changsha, China, and with the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Changsha.

Her research interests include DL and remote sensing image change detection.



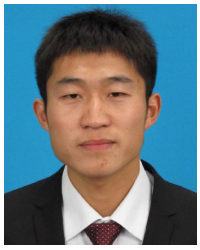
Licheng Liu (Member, IEEE) received the B.S. degree from the China University of Geosciences, Wuhan, China, in 2010, the M.S. degree from Hunan University, Changsha, China, in 2012, and the Ph.D. degree from the University of Macau, Macau, China, in 2016.

He is currently an Associate Professor with the College of Electrical and Information Engineering, Hunan University, Changsha. His research interests include image processing, computer vision, pattern recognition, and machine learning.



Xinxin Liu (Member, IEEE) received the B.S. degree in geographic information system and the Ph.D. degree in cartography and geographic information system from Wuhan University, Wuhan, China, in 2013 and 2018, respectively.

In July 2018, she joined the College of Electrical and Information Engineering, Hunan University, Changsha, China, where she is currently an Associate Professor. Her research interests include image quality improvement, remote sensing image processing, and remote sensing mapping and application.



Yuzhong Ma received the B.S. degree in geomatics engineering from the Shandong University of Science and Technology, Qingdao, China, in 2009, and the M.S. degree in photogrammetry and remote sensing from Peking University, Beijing, China, in 2013.

He is currently working as an Engineer with the Shandong Provincial Institute of Land Surveying and Mapping, Jinan, China. He is mainly engaged in research on remote sensing image processing and remote sensing applications.



Jing Li received the B.S. degree in geomatics engineering from the Shandong University of Science and Technology, Qingdao, China, in 2009, and the M.S. degree in photogrammetry and remote sensing from Peking University, Beijing, China, in 2013.

He is currently working as an Engineer with the Shandong Provincial Institute of Land Surveying and Mapping, Jinan, China. He is mainly engaged in research on remote sensing image processing and remote sensing applications.

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